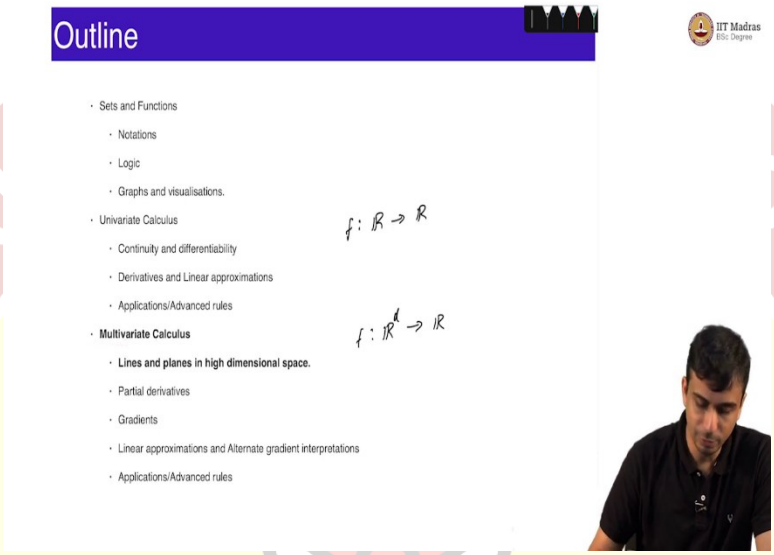


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor Harish Guruprasad Ramaswamy
Department of computer Sciences & Engineering
Indian Institute of Technology, Madras
Multivariate Calculus: Lines and Planes in Higher Dimensional Space

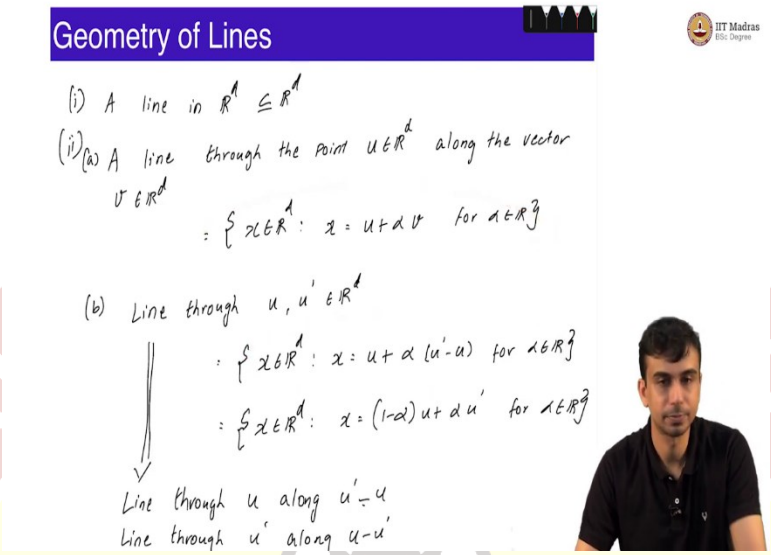
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Hello everyone, and welcome to another lecture of Machine Learning Foundations. We will continue with our calculus recap. We have been seeing about univariate function, univariate calculus, which is about functions f from R to R, the domain and codomain are both R. We will generalize most of these two multivariate functions which are functions from Rd to R, the codomain is still R, the real value, just that the input domain is now d dimensional.

And this brings us lots of intuitive geometric ideas for it, for explaining multivariate calculus ideas. Still, we will be doing only mostly differential calculus and most of these you should have seen in your Mathematics for Data Science 2.

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Before we actually get × the calculus, we should first get the basic geometry of high dimensional space. And the basic sets for doing that are lines and planes. So, what is a line, we will be working with the Rd now, let us, line in Rd is first of all a subset of Rd, line in Rd is a subset of Rd. A line is parameterized by two vectors, a line through the point U in Rd along the direction v in Rd is simply given by the set of all d dimensional vectors such that x can be written as u + α × v for α reals.

We will see some examples soon. But the idea is pretty clear it is a line through the point along the direction is given by this set. Some× you might also see another equivalent version that is a line through two points, it is u and u dash. So, that is another definition this, we will call this definition 2 a and 2 b is a line through the point, through the points u and u dash is simply given by the set of all x in Rd such that x can be written as u + α × u dash - u for α in real. Which is also written as set of all x in Rd such that x = 1 - α× u + α × u dash for α reals.

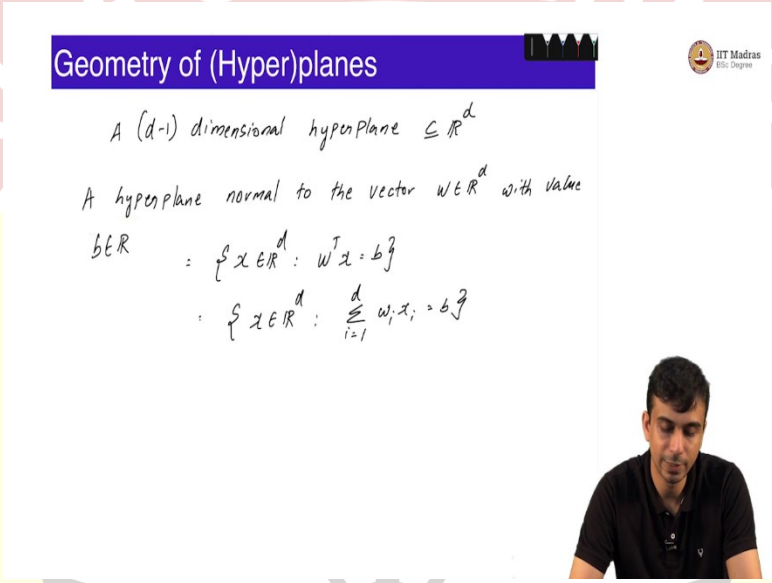
So, as you vary α you get different 1 - α u + α u dash and the union of every such element forms the line. You can see from this expression of the line through u dash and line through u along the vector v that this line through u and u dash is exactly the same as the line through u along the direction u dash - u. Similarly, line through all of these things are clearly the same.

So, this is exactly equal to a line through u along u dash - u. This is also equal to line through u dash along u - u dash. All of these three sets are the same really, line through u and u dash is the

same as line through u along u dash - u, and it is the same as line through u dash along u - u dash, all of these are the same subset of R. You can, this is a simple exercise to prove.

In fact, it is kind of obvious to see from this notation that u + α × u dash - u can just be u dash - u as a vector indicating direction, if you do that, it is exactly the same as this. So that is a line is a subset of Rd. And it does look like line, we will see some examples soon.

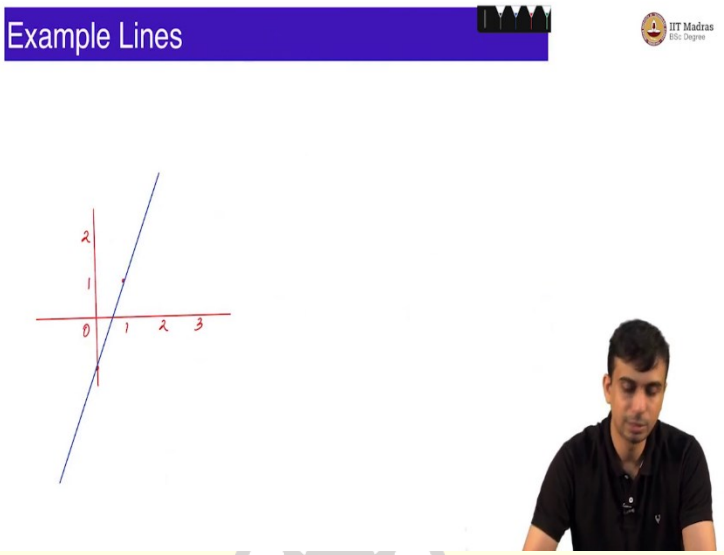
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The next basic set in d dimensional space is a hyperplane or just a plane. A plain, hyperplane is typically a d - 1 dimensional hyperplane. This the default dimension of the hyperplane, this is also a subset of Rd. We will say that a hyperplane normal to the vector W in Rd with value b in R is given by the set of all x in Rd such that W transpose X = b which will set of x in Rd such that the sigma i equal to 1 to d Wi xi = b. So, this the definition of a hyperplane. There are other definitions also. But those are slightly more complicated.

So, this is the standard definition of hyperplane, which is defined in terms of a vector normal to the plane and value b, we will explain what these terms mean soon. But as of now, this is parametrized by a vector w and a value b.

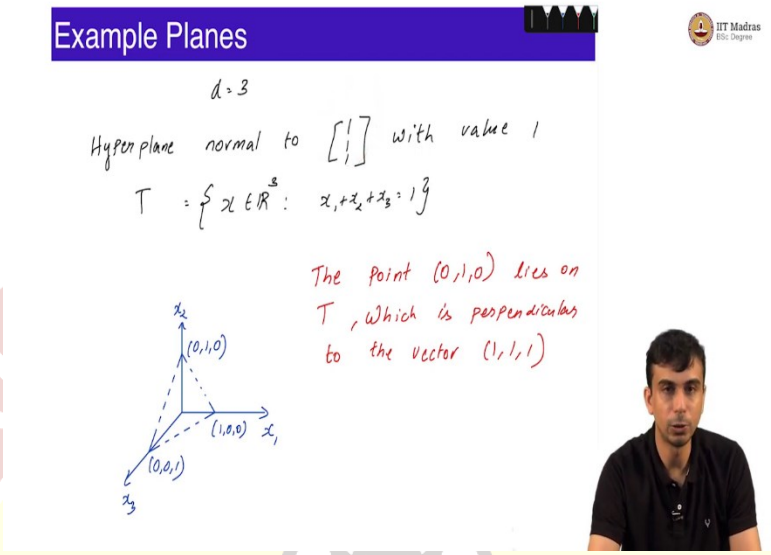
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So, here are a couple of simple examples. For lines first. So, an example line here, this is an example line, a line through consider this set, this a line through the point 1, , 1 along the direction, 1, , 2. So, this would be the set of all x in R2 such that x = 1, 1 + α × 1, 2, where α varies over the real line.

So, visualization would be something like this, it does pass through the point 1, , 1 and it has all scalar multiples of 1, , 1 + α × 1, , 2, all scalar multiples of 1, , 2 can be added to the point 1, , 1. And that would correspond to this blue set. And you can see why it is called a line because it does really look like a line when plotted. So that is a line.

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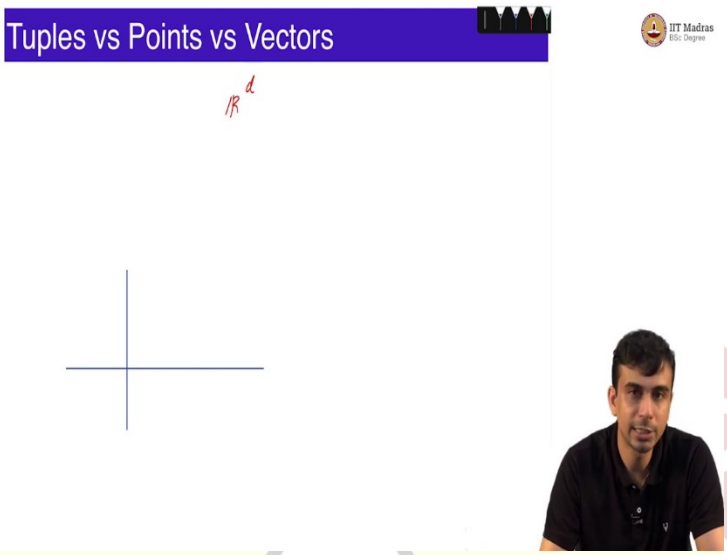
Now we will take a look at some example planes. Here is a simple example plane with three dimensions, d = three, a hyperplane. That is normal to the vector 1, 1, 1, normal, perpendicular, orthogonal are pretty much the same. Normal means perpendicular. Normal with value 1, what is that, it is simply the set of all x in R3 such that x1 + x2 + x3 = 1.

So, visualization of that would, in imagine a 3d, 3 axes given by x1, x2, x3. Here, x1 + x2 + x3 = 1 would correspond to a plane, which a part of it would be this, it could pass through the point 0, 1, 0 1, 0, 0, 0, 0, 1 it will go beyond the, beyond the axis as well. But the visible part, if you can imagine a 3d corner of your room and say imagine a corner of your room a paper, a piece of paper floating there would look like this, corresponding to the hyperplane, would look like this.

So here, we will make a distinction between points, vectors and tuples. So, this particular set, we will call it this the hyperplane T. You can say a statement like this. The point 0, , 1, , 0 lies on T, which is perpendicular or normal, perpendicular to the vector 1, , 1, , 1.

So why is this particular plane called normal or perpendicular to the vector 1, , 1, , 1. Well, what you could do is you could imagine a vector or a ray originating at the origin 0, , 0, , 0 and along 1, , 1, , 1. So that is kind of a ray, that would cut this, that would pierce this plane at 90 degrees. That is the reason why this particular plane is said to be normal, or perpendicular to the vector 1, , 1, , 1.

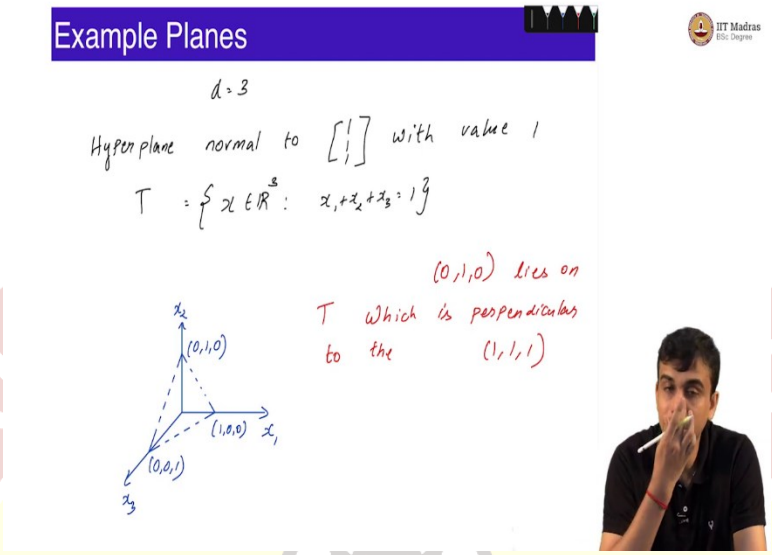
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So, this makes the definition I mean, I have been using vectors, points and tuples interchangeably. Because all of these are represented by Rd. Rd is, a tuple is simply what you would store in, let us say, you are storing a d dimensional vector in a programming language, you would store it in a, as a tuple. So that is, that is how represented, so points and vectors are both represented as tuples.

But they have different meaning though, a point corresponds to, if you take physics, a point corresponds to a location and a vector corresponds to a direction. You can say that I am in Chennai, and I am going, I am going towards Bangalore. So that is a valid, so you can imagine a line through Chennai along the direction, given by Bangalore - Chennai. So that is, the that that is an example line. And directions are essentially vectors and, points are locations. So that is the, there is a difference both of these are represented by a vector in, by tuple in Rd, but there is a difference between point and vectors.

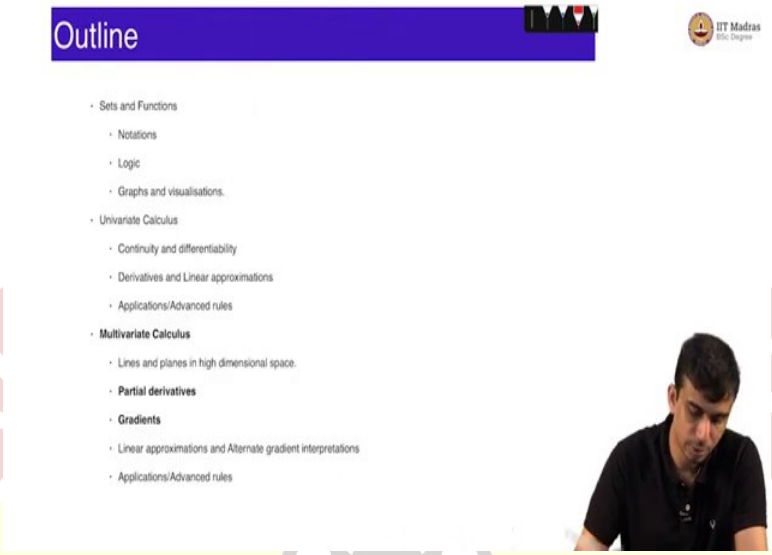
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From context you should be clear whether a tuple is used as a point or as a vector. For example, take this point. So, some× you might not even see, you might say 0, , 1, , 0 lies on T which is perpendicular to 1, , 1, , 1. You can say this and you would not be wrong, even though I have not said the word point here. And the vector here, this would be a perfectly valid statement, because from context, it is clear that 0, , 1, , 0 is used as a point and 1, , 1, , 1 is used as a vector.

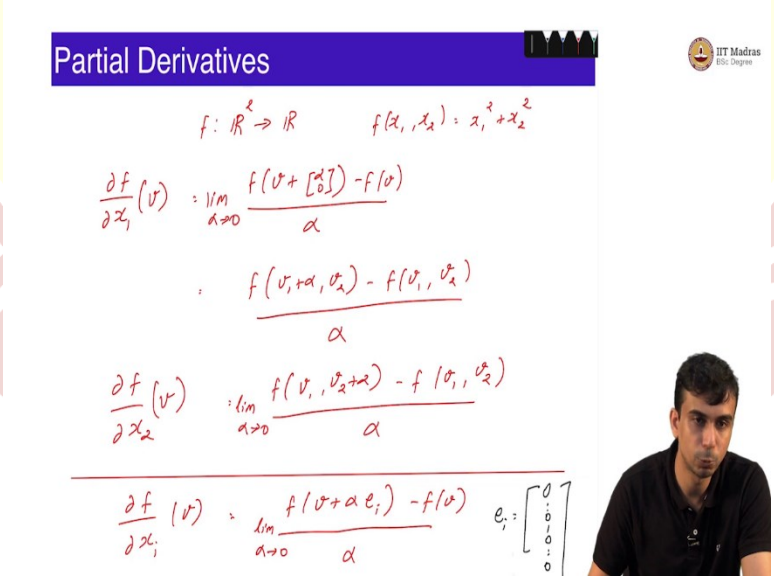
Why is that because 1, , 1, , 1 as a point makes no sense, because you cannot be perpendicular to a point you can be perpendicular to a vector, but you cannot be perpendicular to a point. And similarly, a vector cannot lie on a plane, only points can lie on a plane. But we will be using all of this interchangeably, from context it will be clear whether a particular element in Rd is used as a point or as a vector. Algebraically, it is, algebraically it should to be very clear. But geometrically, you must understand whether a particular vector is, a particular tuple is used as a vector or as a part.

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So, with that, I think that gives the basic geometry tools that are required for understanding multivariate functions. And now we can move on to the basic tool that is, the basic building block of multivariate calculus, which is, which are partial derivatives and gradients.

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So, what is a partial derivative. You have seen this perhaps in your school already. Let us say you can imagine a function f from R2 to R. The simplest multivariate function you can think of which is a two dimensional function. Let us say f of x1, , x2 = x1 square + x2 square, it is an example

function. The partial derivative of a function with respect to let us say one of its arguments, what it means is you view the rest of the arguments as constants, if that is the case, this is still a one dimensional function.

If you fix the other variables, it becomes a one dimensional function, and then you can do the derivative. So that is what the partial derivative means, ∂ f by ∂ x1. So, I am going to use the sound of ∂ to denote this, I might see it called partial or something like that, but I am going to use ∂ because I find it much more convenient than partial, do that. So, ∂ f by ∂ x1 is simply F of ∂ F by ∂ x1 evaluated at a point v.

So, that is simply equal to f of v + I am just going to take d equal to 2 in this case, where we + α 0 - f of v by α. Why is this correct? This is correct because I am fixing the second variable, this is with respect to the first variable x1, so second variable is fixed. So, F of v + α or in other words I will write this even simpler, this is F of v1 + α, , v2 - f of v1, , v2 divided by α.

So, for the purposes of calculation of the partial derivative with respect to the first variable, the second variable is kept constant. So, it is viewed as a constant it is not changing at all, only the first variable is changing. F of v1 + α, and F of v1 + α - F of v1 by α, the second variable is kept constant at v2. So, this is the partial derivative Similarly, you can do the partial derivative of this with respect to the second variable also, what is ∂ f by ∂ x2 evaluated at v. What is that, that is f of v1, , v2 + α - f of v1, , v2 divided by α.

In general, if you have a d dimensional function, how would you do this. If you have a d dimensional function for a general d dimensional function you can still write it this way, that is ∂ f by let us say ∂ xi the ith argument. Let us say f has d arguments x1 to xd ∂ f by ∂ xi evaluated v what would that be, that would be f of v + α × ei - f of v divided by α.

All of this is with limit α approaching 0. What is ei, ei is simply the coordinate vector with the ith coordinate being equal to 1 and the rest being 0. You can easily see that is what is happening here for, to get the first derivative with respect to the first variable you just take v + α × 1, 0, to get the derivative with respect to the second variable you take v + α × 0, 1.

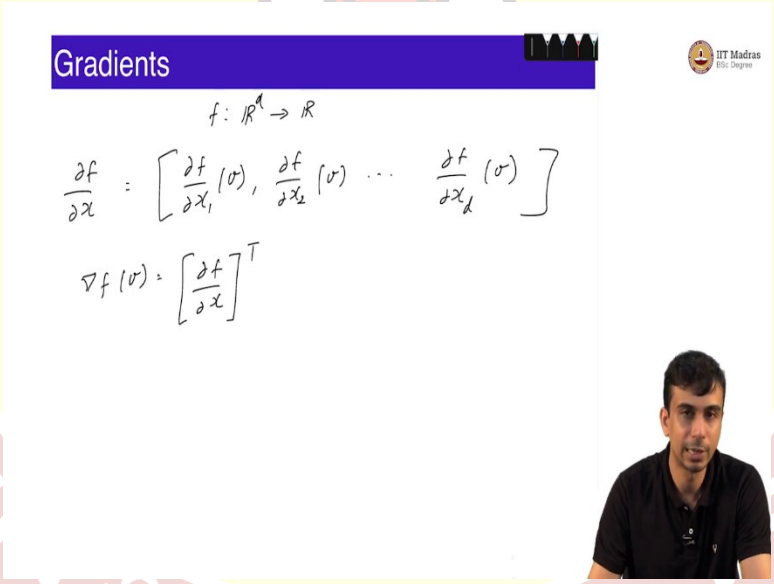
In general, if you have d variables, if you take you want the partial derivative with respect to the ith variable, (())(18:41) × you take f of v + α × ei, where ei is the coordinate vector which has 0,

where ei is a d dimensional vector, it is 0 everywhere else and 1 in the ith position, with there is just a single 1 which is in the ith position.

This is clearly generalizing the case of two variables that we have here. So, all of this, if you had, I mean for the, let us go back to the two variable case and ∂ f by ∂ x1 v. If the second variable is fixed at v2, f is just a function of the first variable, around v1. And in that case, this exactly reduces to the standard derivative expression.

In general, however, this is the general expression for the partial derivative or the, the ith argument you just consider F of v + α × ei - f of v by α, where you let the limit of α approach 0. So, this is for every, we have a d dimensional function, you can evaluate the partial derivative at, you can evaluate the partial derivative of a function at a saddle point, at a point v with respect to the several variables x1 to xd.

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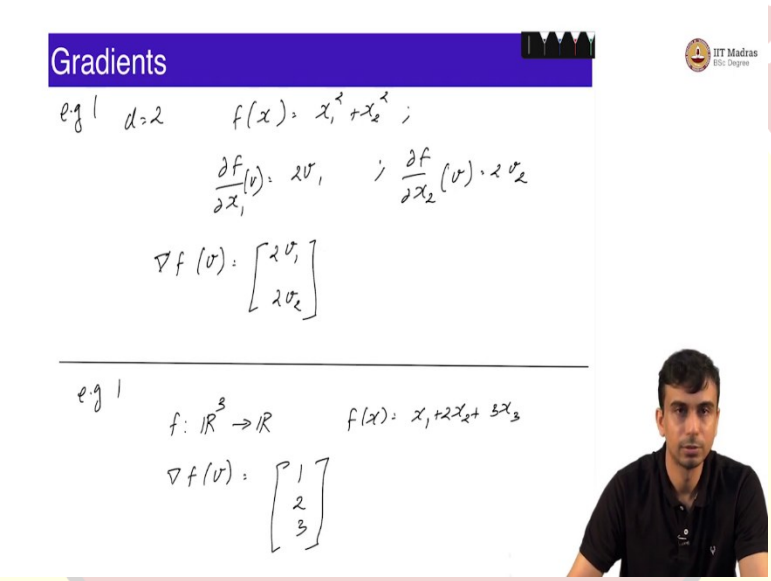
So, now, that means that you have a function f from Rd to R. You have for a given point, for any given point v, you have these things, which is ∂ f by ∂ x1 evaluated at v, ∂ f by ∂ x2 evaluated at v, and so on. ∂ f by ∂ xd evaluated at v. You can imagine putting all of these in a row vector and this you can just call it us ∂ f by ∂ x.

So, this f is a function, this x is a vector. So that we can take a derivative of a function with respect to a vector. By that we mean this particular vector, where you take you package in all your partial

derivatives × one row vector. Often, you would also use a gradient, which is a gradient of f at v is simply ∂ f by ∂ x transpose.

Why is that, because the gradient is typically written as a column vector. But derivative of f with respect to x, where x is a d dimensional vector is written as a row vector. Otherwise, it is the same thing, it is the gradient is also a collection of partial derivatives, it is the d dimensional vector which collects partial derivatives.

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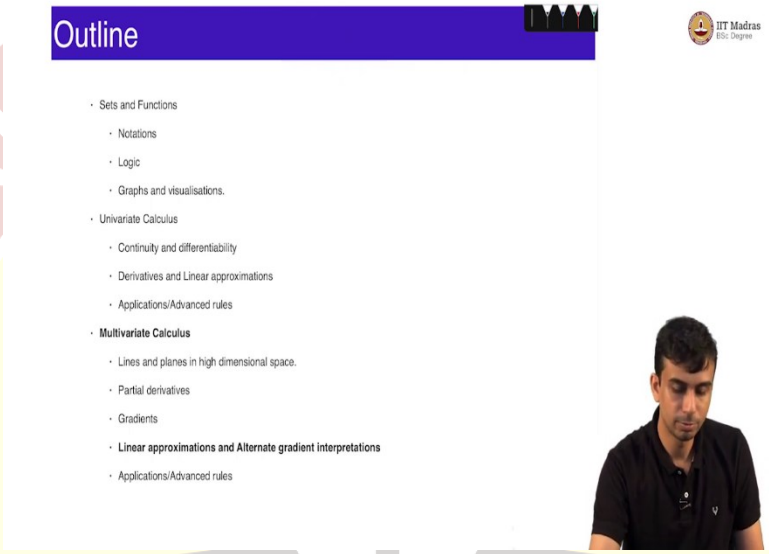
So, we can have some simple functions and take the derivatives. Let us use the same function which is f of x = x1 square + x2 square. In this case, let us say d = 2, it is a 2 dimensional function, f of x equal x1 square + x2 square. What is ∂ f by ∂ x1, well, that is simply equal to 2 x1 ∂ f by ∂ x2 is simply equal to 2x 2. ∂ f by ∂ x1 evaluated at v is simply 2 v1, 2 v2, so ∂ f, I think it is better to write ∂ f by x1 evaluated at v = 2 v1, ∂ f by ∂ x2 evaluated at v is equal 2 v2.

So, the gradient of f evaluated v is a column vector given by 2 v1 2 v2, you can have other functions, we will have. This is the first example. Another example might be, let us say you have a function f from R3 to R as follows. F of x = let us say x1 + 2x 2 + 3x 3. What is the gradient of f with respect to, what is the gradient of f evaluated at some point v? Well, that is the vector given by ∂ f by ∂ x 1, ∂ f and ∂ f by ∂ x2 ∂ f by ∂ x3.

And if you take the, if you view x2 and x3 as constants, and you take the derivative of f with respect x1, you get simply 1. Similarly, if you take the derivative with respect to the second

variable is x 2, you get 2, you get for the third variable you get 3. So, the gradient happens to be a constant. And these kind of functions where the gradient is a constant is called a linear function. f of x equals x 1 + 2x 2 + 3x 3 it is linear in all its arguments. And that is the reason why the gradient is a constant.

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So, with that, I think we have seen the basic tool, the basic partial derivatives and gradients that are required for multivariate calculus. We will see some more applications of these gradients and partial derivatives soon. But we will wrap this video for now.